



Cross-Linguistic Inequalities in Nigerian Political Knowledge across Wikipedia and Wikidata: A Computational Audit and Controlled RAG Experiment

GROUP 3

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Background

Two linked but different layers of the Wikimedia knowledge ecosystem



Wikipedia

Language-specific encyclopedic narratives. Each language edition is written and curated independently — coverage, depth, and framing can differ from one language to the next.

Distinct language editions · independent narratives



Wikidata

One shared structured knowledge graph of entities and relationships, with language-dependent labels, descriptions, aliases, and links back to Wikipedia articles.

One shared graph · language-dependent labels

Prior research shows Wikimedia knowledge is unevenly represented across languages, regions, and social groups — and political knowledge is especially sensitive.

Problem Statement

A Nigerian political actor or party may have a detailed English Wikipedia article, but limited coverage in a Nigerian language — while its Wikidata item may have labels, descriptions, or references missing in some languages altogether.



Cross-language gaps

Missing articles, absent or limited Wikidata labels and descriptions across languages.



Weak sourcing

Incomplete statements and inconsistent reference coverage between languages.



Framing differences

Different narrative emphasis on the same political entity, language to language.



Uncertain AI impact

Multilingual AI differences can't automatically be blamed on Wikimedia — the model itself may carry pre-existing bias.

Aim of the Study

To quantify and compare political bias between English and selected Nigerian-language Wikimedia projects, in order to identify systemic neutrality gaps in regional political coverage.



Assess

How political information varies across languages



Compare

Wikipedia vs. Wikidata consistency and completeness



Test

Whether the gaps measurably affect AI outputs

Specific Objectives

Three steps guide the study, from measurement to downstream AI impact

01

Assess variation

Examine how information about Nigerian political figures and institutions varies across languages on Wikipedia and Wikidata.

02

Test systematic patterns

Determine whether differences vary systematically by language, entity type, and salience — and check consistency between Wikipedia and Wikidata.

03

Trace AI impact

Test whether identified Wikimedia asymmetries measurably influence multilingual, knowledge-grounded AI outputs; derive indicators for continuous auditing.

Research Questions



RQ1

How does information about Nigerian political figures and institutions vary across languages on Wikipedia and Wikidata?



RQ2

Are observed differences systematically associated with language, political-entity type, or entity salience — and how consistent are Wikipedia and Wikidata representations of the same entities?



RQ3

Do Wikimedia knowledge asymmetries produce measurable differences in multilingual AI outputs when equivalent political queries are tested under controlled retrieval conditions?

What the Literature Shows

Wikimedia projects are multilingual knowledge infrastructure — but the evidence base is uneven



Cultural imbalance

Large content imbalances found across 40 Wikipedia language editions (Miquel-Ribé & Laniado, 2018).



Demographic bias

Race and citizenship disparities identified in Wikidata (Shaik et al., 2021); bias varies by context (Das et al., 2023).



Coverage & precision

Non-English entity names and descriptions show lower coverage and precision in multilingual knowledge graphs (Conia et al., 2023).



Political distortion

Notable over- and under-representation of political parties across DBpedia and Wikidata (Karadeniz et al., 2022).



Provenance gaps

Quality and accessibility of Wikidata sources vary by language (Amaral et al., 2021).



Model-level effects

Cross-language differences in political ideology found in multilingual LLMs (Nadeem et al., 2026).

Research Design

A comparative cross-sectional computational audit, with a controlled experimental component



Component 1 — The Audit

Observational & comparative

Audits how selected Nigerian political entities are represented across relevant Wikipedia language editions and Wikidata — coverage, completeness, provenance, and framing.

Nigeria's national political entities · English + selected Nigerian languages



Component 2 — The AI Test

Controlled & experimental

A retrieval-augmented generation (RAG) experiment tests whether the Wikimedia asymmetries found in Component 1 influence downstream, multilingual AI outputs.

Matched multilingual queries · no-retrieval, raw, and harmonised evidence conditions

Data Sources & Instruments

SOURCES OF DATA



Wikidata

IDs, labels, descriptions, aliases, sitelinks, statements & references



Wikipedia

Article presence, lead-section text, metadata across languages



External benchmarks

Electoral, parliamentary & government records



AI experiment outputs

Responses to matched multilingual political queries

FOUR INSTRUMENTS

1

Entity-sampling & benchmark matrix

Role, electoral relevance, Wikidata ID, available language pages

2

R scripts & SPARQL/API templates

Reproducible extraction into structured datasets

3

Structured audit checklist

Scores presence, completeness, reference coverage, consistency

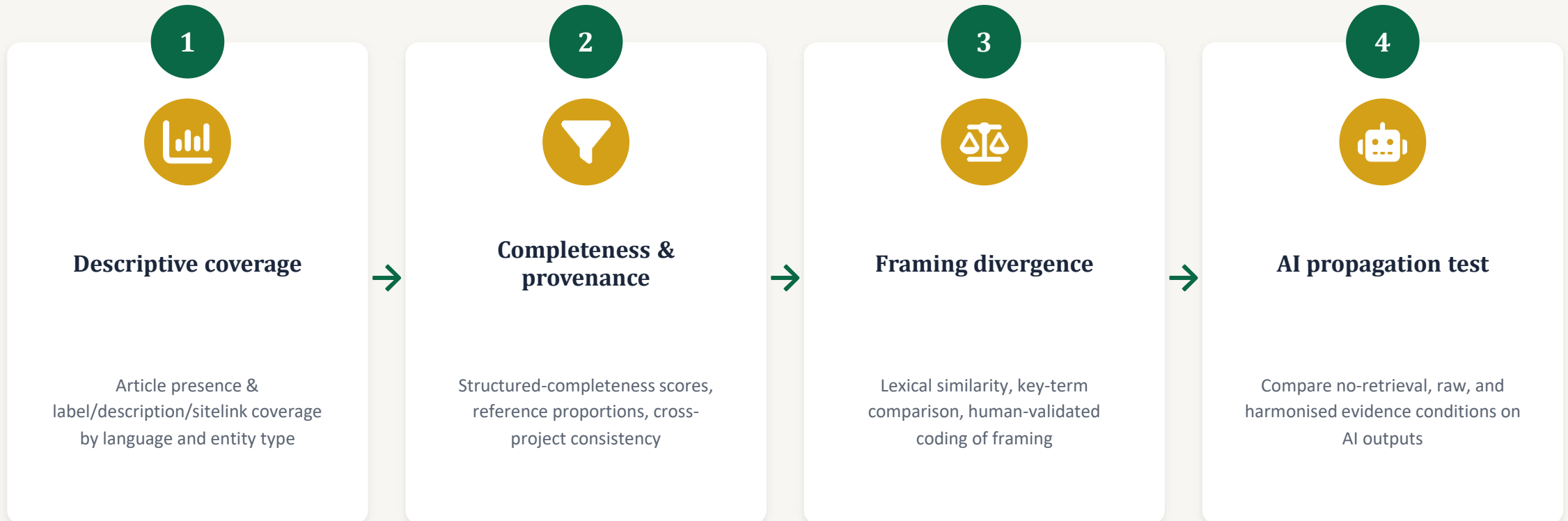
4

Framing-analysis codebook

Classifies omission, ideological wording, emphasis, contradiction

Analysis in Four Stages

R is the principal environment throughout — extraction, cleaning, analysis and visualisation



Expected Contribution & Significance



A reproducible framework

For auditing political knowledge across the linked Wikimedia ecosystem, Nigeria-focused.



Distinguishes the layers

Separates Wikipedia language-edition differences from Wikidata's shared structured layer.



Tests AI impact

Shows whether Wikimedia gaps can affect knowledge-grounded AI responses.



Practical indicators

An R-based workflow and audit indicators for Wikimedia communities, researchers, and AI developers.

Findings will target publication in a high-indexed, peer-reviewed Scopus journal — contributing to computational social science from the Global South, with a Nigerian perspective.

Anticipated Limitations

Honest boundaries on what the study can — and cannot — claim



Framing is contested

Cross-language differences are not automatically evidence of bias.



Sparse content

Some Nigerian-language Wikimedia content may limit statistical power — though absence is itself a finding.



A moving target

Wikimedia content changes continuously; the study uses a fixed, time-bounded snapshot.



Bounded generalisation

The downstream AI test cannot be generalised to every commercial system or hidden training corpus.



Thank You

Questions & Discussion